Universal Motives: Fear, Food & Music

An Evolutionary Perspective on Motivation

This lecture explores three fundamental motives that shape human behavior, each with deep roots in our evolutionary past. We will examine:

- **Fear**: How did fear evolve as a primary survival mechanism, and what stimuli are we innately prepared to avoid?
- **Food**: What governs our innate preferences for some tastes and aversions to others, and how does early experience shape them?
- **Music**: What is the evolutionary purpose of music—a complex, universal human behavior with no obvious survival advantage?

Each topic will build upon the last, progressing from clear-cut adaptations to more nuanced and mysterious evolutionary questions.

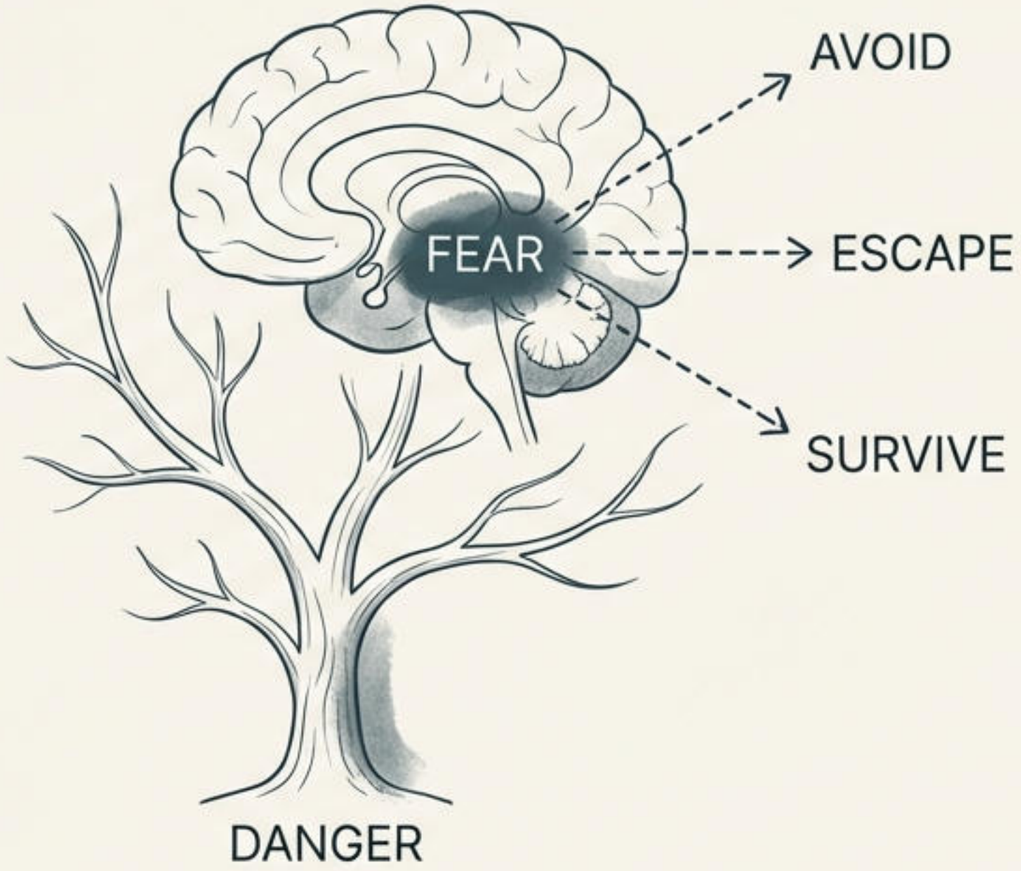
NotebookLM

Fear & Survival

The Adaptive Value of Fear

Fear and anxiety are not mere discomforts; they are highly adaptive emotional states. Their primary evolutionary function is to motivate avoidance and escape behaviors, which are critical for coping with environmental dangers.

Many of the stimuli that evoke fear are not products of individual learning or traumatic experiences. Instead, they appear to be remnants of our evolutionary history, passed down because they conferred a survival advantage to our ancestors. This raises a central question: what are we innately predisposed to fear, and why?



NotebookLM

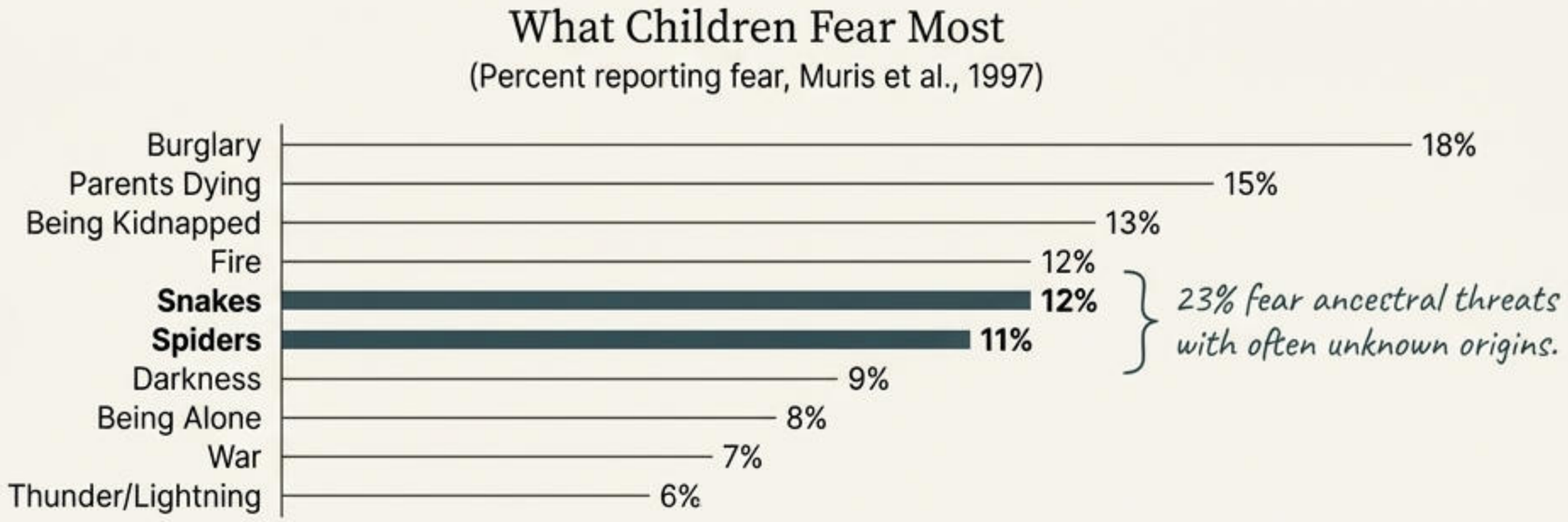
Common Fears: An Evolutionary Echo

Surveys across cultures reveal a consistent pattern in our greatest fears, many relating to ancestral threats rather than modern dangers.

- **Dutch Children (9-13):** Top fears included being burglarized, but also snakes and spiders.
- **Australian Children (11-14):** Duplicated many fears, including parents dying, being hit by a car, but also snakes, the dark, and spiders.
- **American Adults:** When choosing from a list, snakes and spiders were prominently selected.

The Mystery of Origin

A significant percentage of individuals cannot recall any specific conditioning event that initiated their fear. Muris et al. (1997) found that 46% of children with a fear of spiders and 33% overall did not know how their fears began, suggesting an innate predisposition rather than a simple learned response.



NotebookLM

Survival Functions & Neural Substrates

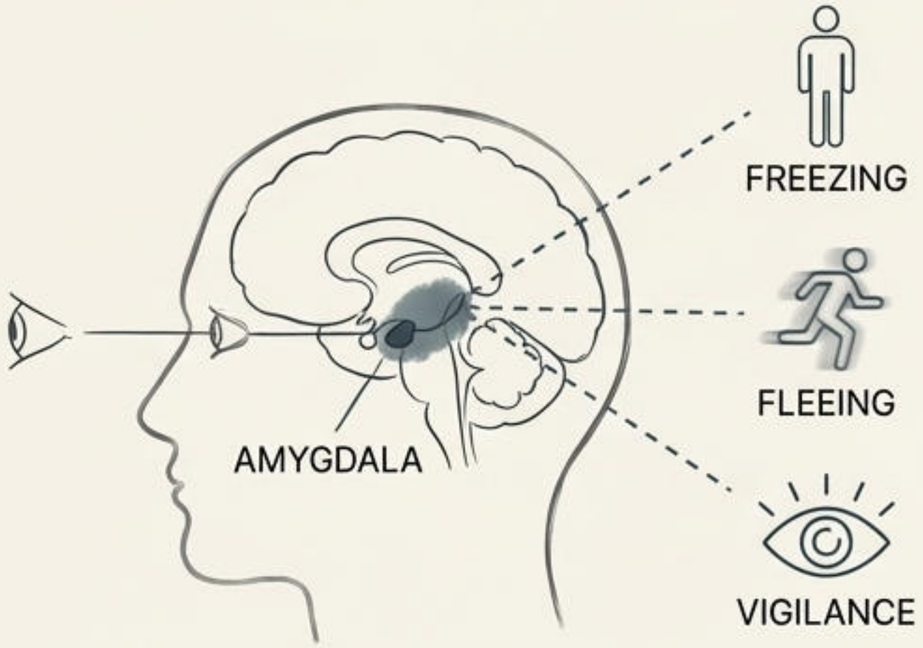
Survival Functions & Neural Substrates

Evolutionary theory posits that fear evolved to help animals cope with imminent danger in two primary ways:

1. **Preparation for Action**: Fear motivates physiological and behavioral readiness for escape or avoidance. This can manifest as freezing, scrambling for cover, fleeing, or tonic immobility, depending on the species and context.
2. **Specific Stimulus Fear**: We have evolved to fear specific stimuli that represented recurrent threats. Öhman (1986) suggests the prevalence of reptile fear may be an archaic remnant from a time when mammals were prey for large reptiles.

The Brain's Threat Detector

Öhman (1993) speculates that natural selection favored the development of neural **feature detectors** sensitive to the stimulus characteristics of ancient predators (e.g., the curvilinear shape of a snake). These detectors may be located in the **amygdala**, a brain region highly sensitive to ambiguous or potentially threatening stimuli.



NotebookLM

Innate Threat Detection: Evidence from Childhood

The ability to rapidly detect evolutionarily relevant threats appears early in life, long before extensive learning could occur.

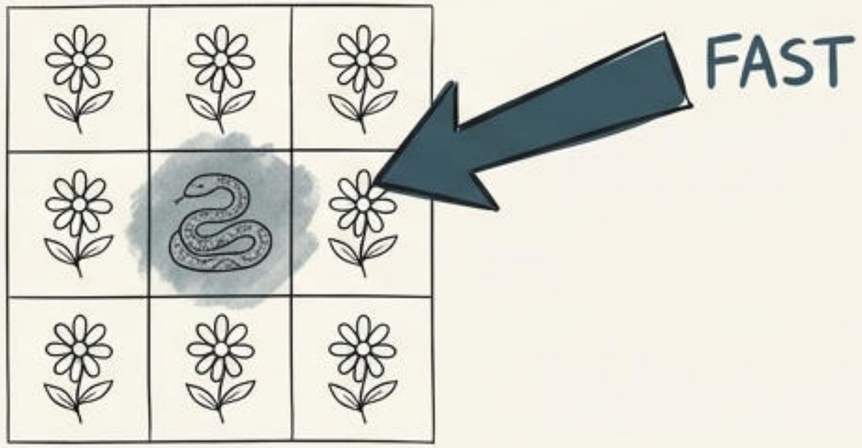
The Snake in the Flowers Study

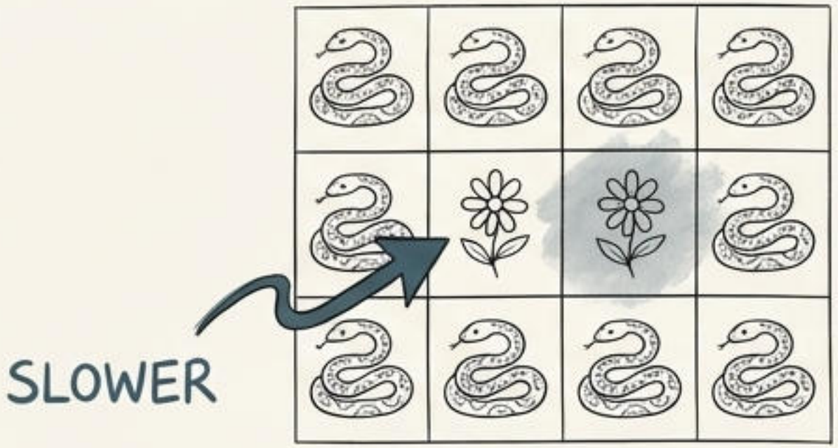
LoBue and DeLoache (2008) tested this by presenting 3- to 5-year-old children with 3x3 matrices of images. Children were asked to find the single target image among eight distractors.

- Task 1: Find a snake among 8 flowers, frogs, or caterpillars.
- Task 2: Find a flower, frog, or caterpillar among 8 snakes.

Results

Children located the snake target significantly faster than any other non-threatening target. This suggests a perceptual **bias** or attentional priority for fear-relevant stimuli, indicating an innate detection mechanism.





NotebookLM

Threat Detection in Infancy: The Role of Motion and Sound

This attentional bias is present even in infancy and is tuned to specific cues.

The Frightened Voice Study

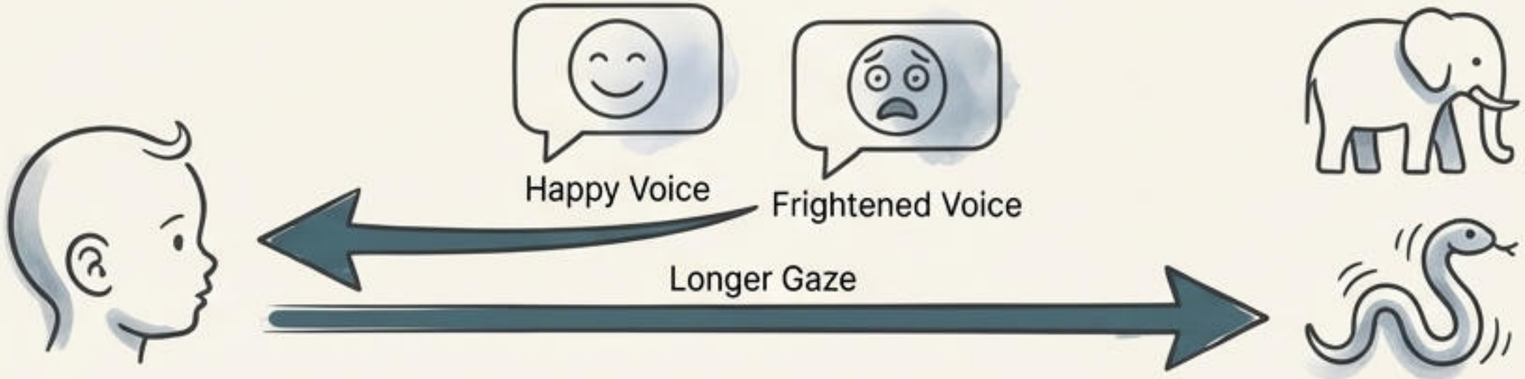
DeLoache and LoBue (2009) showed 7- to 18-month-old infants pairs of moving animal videos (e.g., a slithering snake and a moving elephant) simultaneously. While watching, infants heard nonsense words spoken in either a happy or a frightened voice.

- Hypothesis: A frightened voice signals danger, which should direct attention toward the evolutionarily relevant threat.

Results:

- Frightened Voice: Infants looked significantly longer at the snake.
- Happy Voice: Infants looked at both animals equally.

Crucial Detail: This effect only occurred when the snake was depicted with its characteristic slithering motion. Infants showed no attentional preference for stationary snakes, suggesting the detection mechanism is highly tuned to specific, predictive features of the threat.



NotebookLM

Preparedness Theory: An Instinct "Lurking"

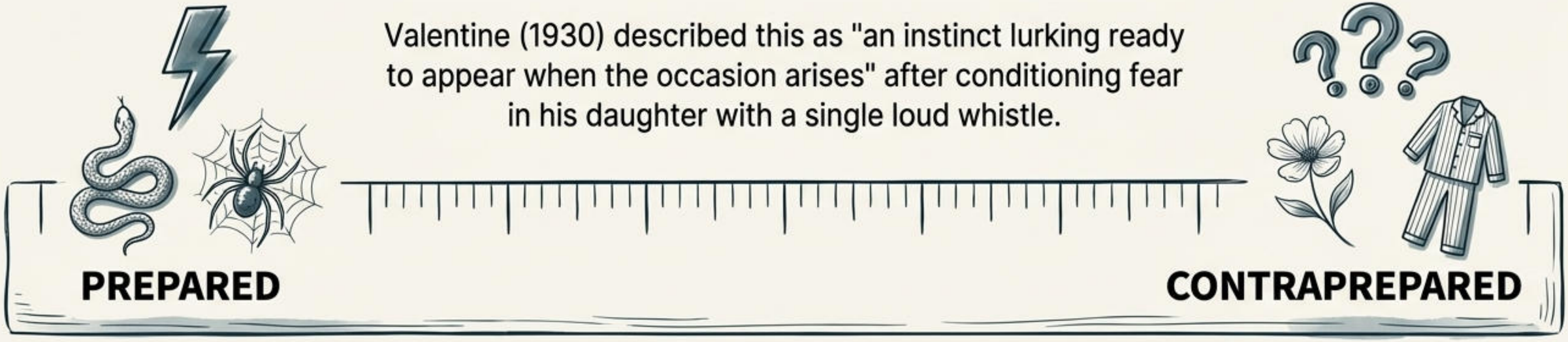
While some fears seem innate, learning still plays a role. However, we are not equally capable of learning all fears.

Seligman's (1971) Concept of Preparedness

Seligman proposed that evolution has "prepared" us to learn certain associations more easily than others. This is not about innate fear, but an innate readiness to *become* afraid of certain things.

- **Prepared Behaviors:** Easily learned, often after a single trial. We are prepared to fear stimuli that were dangerous in our ancestral environment (snakes, spiders, heights, dark).
- **Contraprepared Behaviors:** Learned with great difficulty, if at all. We are contraprepared to fear stimuli that are modern but dangerous (electrical outlets, hammers) or ancestrally benign (flowers, pajamas).

Valentine (1930) described this as "an instinct lurking ready to appear when the occasion arises" after conditioning fear in his daughter with a single loud whistle.



NotebookLM

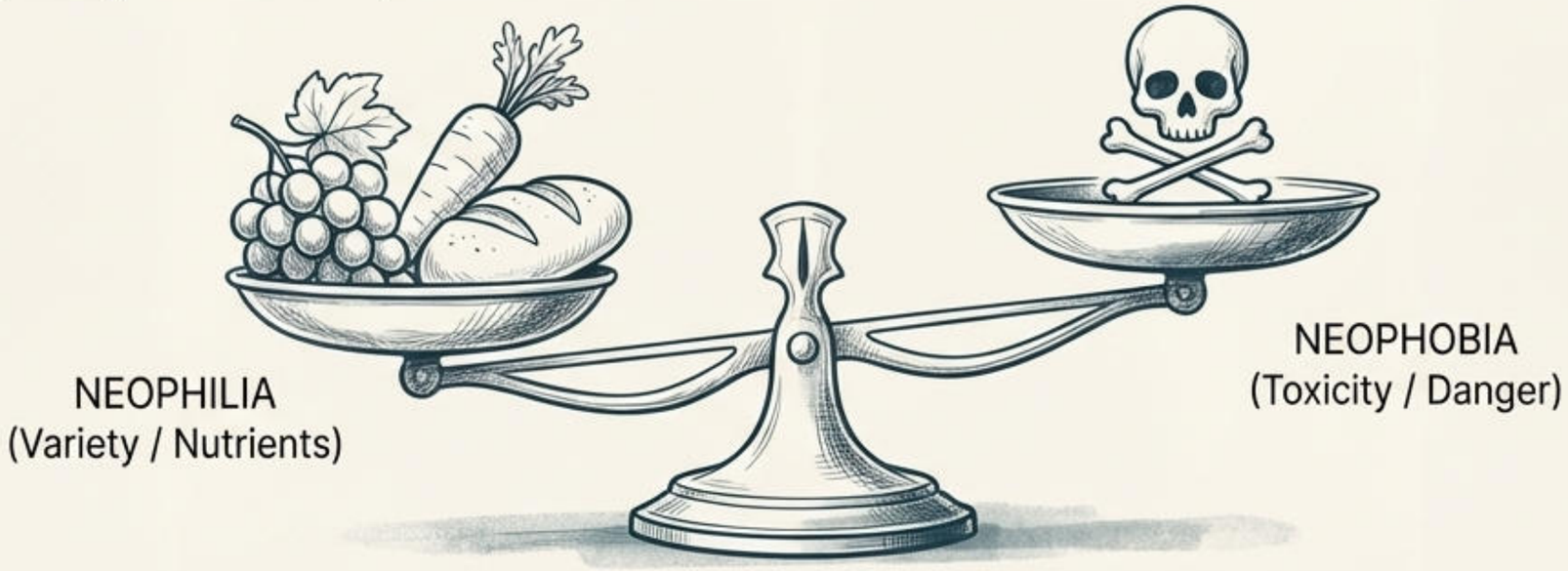
Food Preferences

An Omnivore's Dilemma

Like fear, our preferences for food are universal motives shaped by evolution. Humans are omnivores, capable of eating a wide variety of foods. This flexibility allowed our ancestors to adapt to diverse geographical locations and food sources. However, this flexibility comes with a risk: the danger of consuming toxins or spoiled substances.

Our food preferences and aversions represent an evolved solution to this dilemma, balancing the need for novel nutrients (**neophilia**) with the need to avoid poison (**neophobia**).

To understand the innate basis of these preferences, researchers often study young children, whose tastes are not yet fully dre shaped by cultural and parental influence.

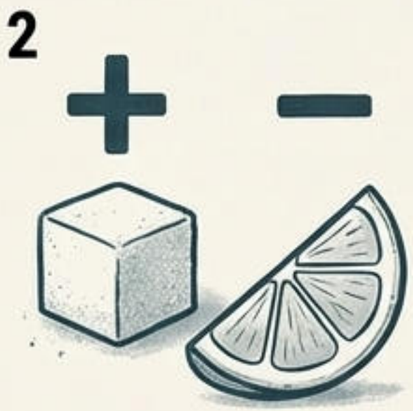


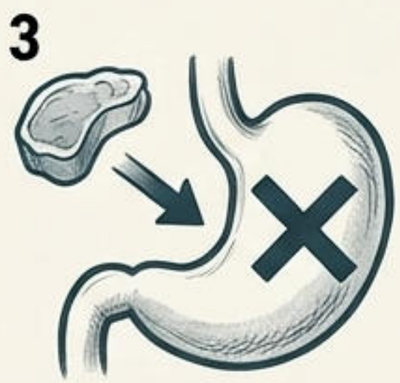
NotebookLM

General Principles of Innate Food Preferences

Research with infants and young children has revealed several core dispositions regarding food:

1. **Food Neophobia:** A tendency to avoid novel foods and prefer familiar ones, particularly evident when children transition from milk to solid foods.
2. **Innate Taste Preferences:** An innate preference for sweet tastes and, later, salty tastes. Conversely, there is an innate avoidance of sour and bitter tastes.
3. **Prepared Learning:** Children are biologically prepared to rapidly form aversions to foods associated with negative digestive consequences (e.g., nausea).
4. **Fat Preference:** Humans show a preference for fatty foods, likely due to their high caloric density, which was valuable in an ancestral environment of food scarcity.





NotebookLM

Measuring Neophobia: Willingness to Try Novel Foods

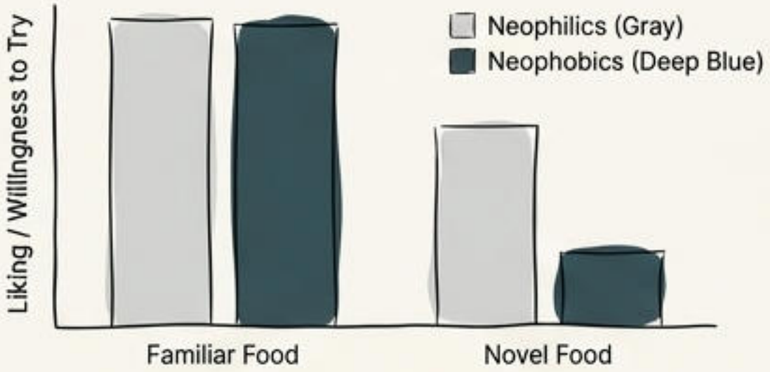
The universal tendency of food neophobia varies between individuals. **Pliner** and **Hobden** (1992) developed the **Food Neophobia Scale** to classify individuals as:

- **Neophobics**: High scorers who strongly avoid new foods.
- **Neophilics:** Low scorers who are willing to try new foods.

Experimental Evidence (Raudenbush & Frank, 1999)

Researchers presented both groups with familiar foods (e.g., yogurt, applesauce) and unfamiliar foods (e.g., hummus, halvah).

Results:
While both groups rated **novel foods less favorably** than familiar ones, the effect was dramatically stronger for **neophobics**. They were less willing to try novel foods, expected to like them less, **actually liked them less, ate less**, and were **less willing to try them again**.



NotebookLM

The Survival Function of Our Senses

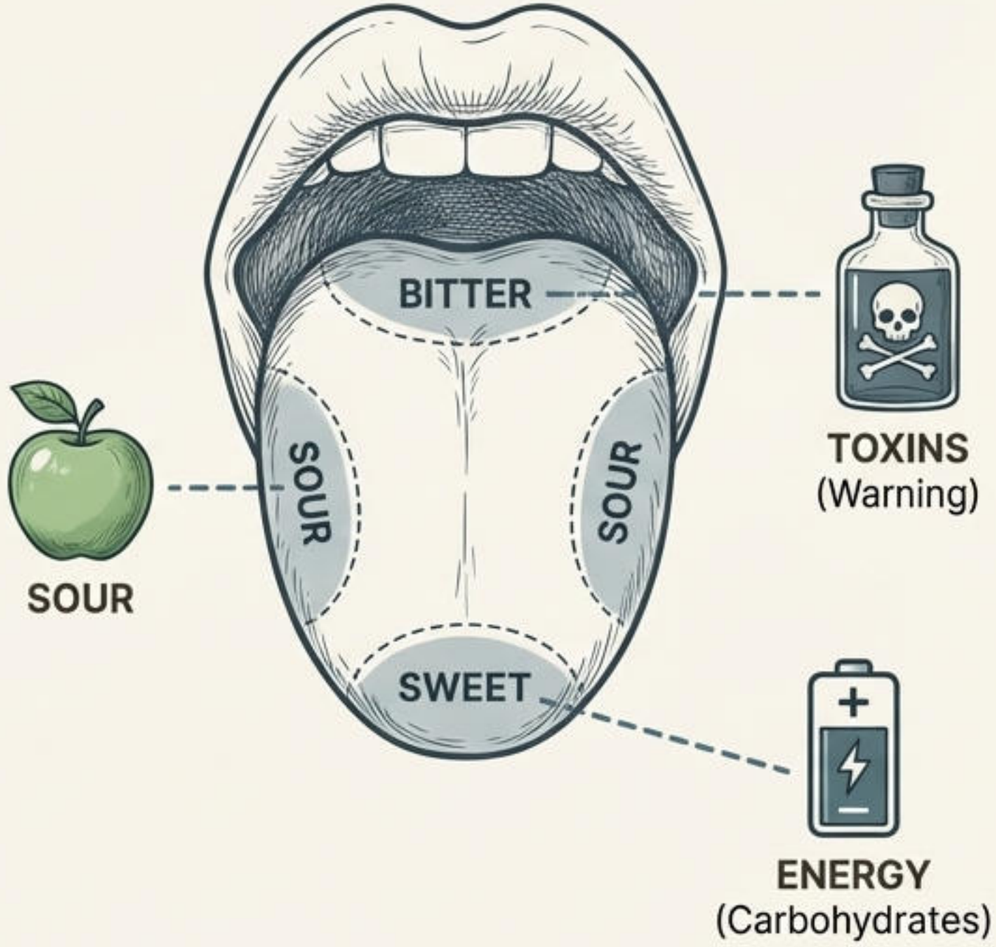
Neophobia's Protective Role

The primary evolutionary explanation for neophobia is that it serves as a protective mechanism. By being cautious about unfamiliar foods, an individual reduces the risk of ingesting a natural toxin, which could cause illness or death. This tendency diminishes with repeated safe exposure to new foods.

Innate Tastes as Chemical Signals

Our innate preferences and aversions for basic tastes function as reliable guides to the chemical properties of potential foods.

- **Sweet**: Signals the presence of sugars (carbohydrates), a primary source of energy. Found in ripe fruits and mother's milk.
- **Sour**: Often indicates unripe fruit or spoilage from bacterial action.
- **Bitter**: Strongly associated with the presence of potentially toxic alkaloids in plants. Serves as a warning to avoid bad food.



NotebookLM

Evidence from the First Hours of Life: Steiner's Studies

The innate nature of taste responses is powerfully demonstrated by observing newborns before their first feeding.

Steiner's (1977) Facial Expression Research

Steiner placed flavored liquids on the tongues of infants, observing distinct, universal facial reactions:

- **Sweet Stimulus:** Elicited a smile-like retraction of the mouth corners, sucking, and licking of the upper lip.
- **Sour Stimulus:** Caused lip pursing, nose wrinkling, and eye blinking.
- **Bitter Stimulus:** Produced a flat tongue protrusion, spitting, and an expression of dislike or anger.

Evidence from Anencephalic Infants

Crucially, Steiner tested **infants born without a cortex**, possessing **only a brain stem and midbrain**. These infants showed the same stereotypical taste responses. This finding strongly suggests the reactions are not learned but are reflexive, controlled by lower brain structures, and are truly innate.

Sweet Sour Bitter

NotebookLM

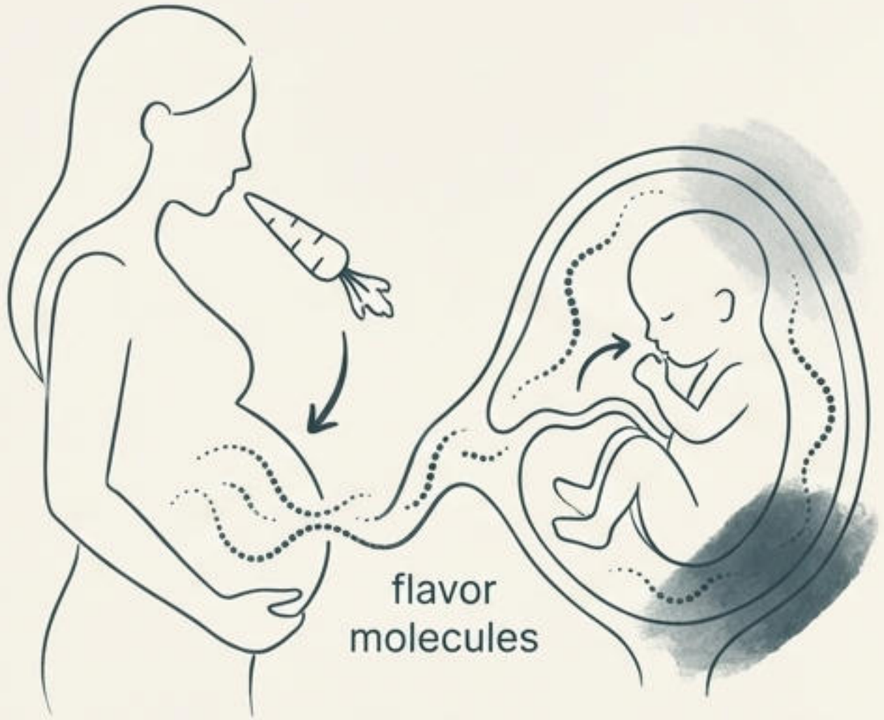
Prenatal Learning: Flavor Experience in the Womb

While basic taste preferences are innate, specific flavor preferences can be shaped by experience even before birth.

Amniotic Fluid as a Flavor Bridge

Flavors from the mother's diet can be transmitted to the amniotic fluid, which the fetus regularly swallows. This exposes the developing fetus to the chemical signatures of the foods common in its mother's (and therefore its culture's) diet.

This early exposure serves a critical adaptive function: it prepares the infant to accept the foods of its culture, smoothing the difficult transition from milk to solid foods. Research by Mennella and Beauchamp (1998) shows that newborns can even detect and prefer the odor of their own amniotic fluid, demonstrating the power of this early learning.



NotebookLM

The Carrot Juice Study: Empirical Evidence for Prenatal Learning

Mennella et al. (2001) designed an experiment to test the effects of prenatal and early postnatal flavor exposure.

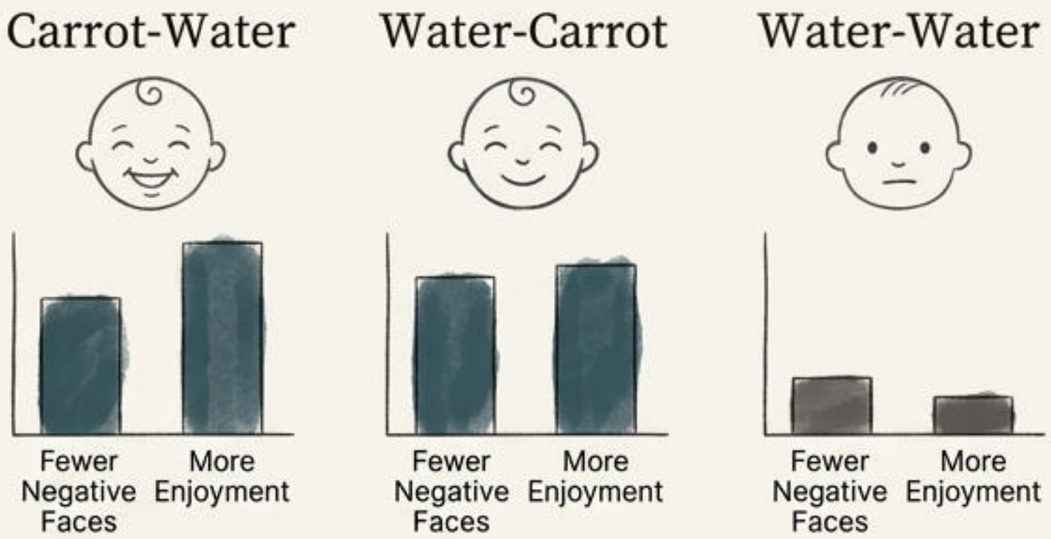
Experimental Design:

Three groups of mothers were assigned different beverage regimens:

- **Carrot-Water**: Drank carrot juice during pregnancy, then water while breastfeeding.
- **Water-Carrot**: Drank water during pregnancy, then carrot juice while breastfeeding.
- **Water-Water**: Drank water during both periods (control group).

Testing & Results:

When infants began eating solid foods, those exposed to carrot flavor either prenatally or via breast milk showed significantly greater acceptance of carrot-flavored cereal **(fewer negative faces, higher maternal enjoyment ratings)** compared to the control group. This confirms that very early flavor experiences shape later food preferences.



(Further evidence: Hepper et al., 2013, found that 8-9 year-old children of mothers who ate garlic during pregnancy ate more garlic-flavored potatoes than controls.)

NotebookLM

The Evolutionary Puzzle of Music

We now turn from clear survival motives to a more enigmatic universal behavior: **music.**

Darwin's Mystery

Charles Darwin himself was perplexed by music's evolutionary origins. In *The Descent of Man*, he wrote that the capacity to produce and enjoy music is not "of the least use to man in reference to his daily habits of life," ranking it among the "most mysterious" faculties with which humans are endowed.

Defining Music and Entrainment

For our purposes, music refers to organized sound patterns that are regular and periodic. A key feature distinguishing music from speech is its capacity to induce **entrainment—the synchronization of body movements** (like tapping or **dancing) to an external auditory beat.** The pleasure we derive from this is obvious, but its **evolutionary function** is not.

"...ranked among the most mysterious with which he is endowed."

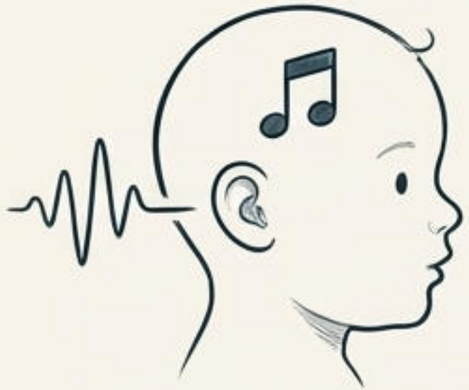
NotebookLM

The Case for Music as Part of Human Nature

While its survival function is unclear, several lines of evidence suggest music is a fundamental, evolved aspect of human psychology (Peretz, 2006).

1. **Ancient Origins:** Music is not a recent cultural invention. The discovery of bone flutes indicates music has been part of human culture for at least 30,000 years.
2. **Universality:** While forms vary, music is present in every known human culture.
3. **Innate Response:** The human response to music appears to be innate. Infants respond to music differently than other sounds, long before any formal training.
4. **Source of Pleasure:** Music is a reliable source of pleasure for the vast majority of people, motivating its creation and consumption.

These facts suggest music is not a trivial pastime but a core component of the human experience that requires an evolutionary explanation.



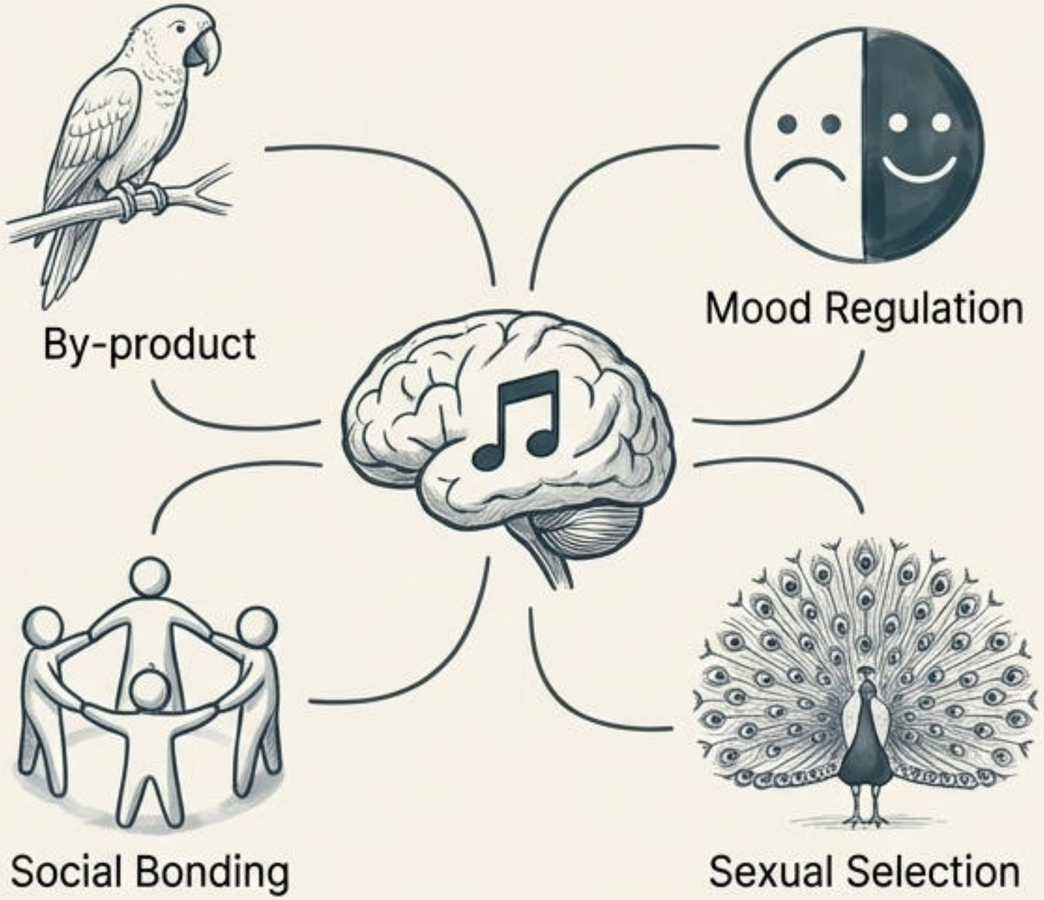
NotebookLM

Hypotheses for the Evolution of Music

If music does not directly aid survival, how might it have evolved? Several major hypotheses have been proposed:

- **A By-product of Another Adaptation**: Music may be an evolutionary side effect. The capacity for entrainment might be a by-product of vocal mimicry—the ability to imitate sounds, crucial for language. The fact that some vocal-mimicking birds (like cockatoos) can entrain supports this.
- **Mood Regulation**: Music reliably arouses emotions. This property could have been selected as a tool to alleviate negative moods and enhance positive ones, improving psychological well-being.
- **Social Bonding**: Group musical activities (singing, dancing) create powerful feelings of cohesion. This could have been adaptive by strengthening group bonds critical for survival.

- **Sexual Selection**: Darwin's preferred explanation. Music may have evolved as a courtship display to "charm" potential mates, signaling intelligence, creativity, and health, thus increasing reproductive success.



NotebookLM

Origins of Musical Taste

Consonance vs. Dissonance: Innate or Learned?

A key question is whether our aesthetic preferences are hardwired or culturally acquired. The most basic preference in Western music is for **consonance** (harmonious, pleasant-sounding note combinations) over **dissonance** (clashing, unpleasant-sounding combinations).

The Nativist Hypothesis

Early research seemed to support an innate preference. Zentner & Kagan (1998) found that four-month-old infants preferred listening to consonant over dissonant music, suggesting this preference exists prior to significant cultural experience. If our preference for harmony is innate, it could point to a deep biological basis for music appreciation. However, more recent evidence has challenged this conclusion.

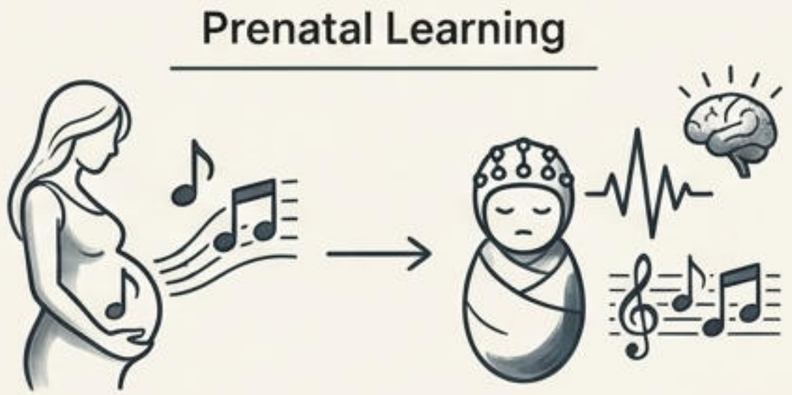
NotebookLM

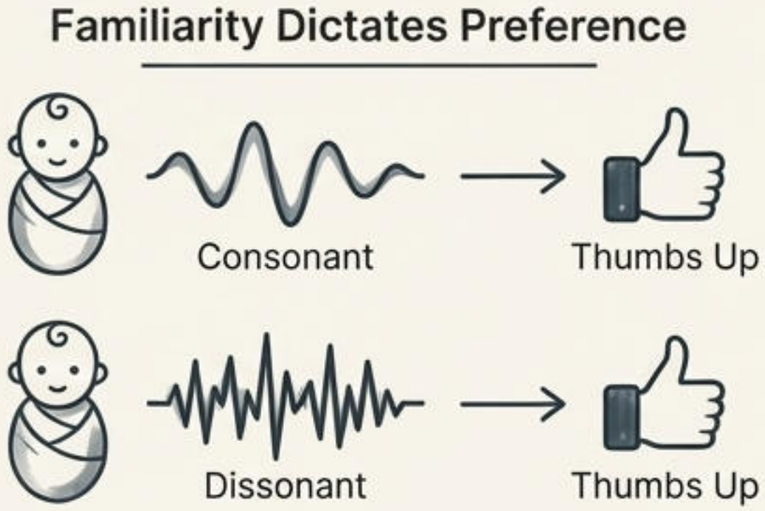
Challenging Innateness: The Power of Familiarity

Two lines of research suggest musical preference is heavily influenced by early exposure, not innate wiring.

- **1. Prenatal Learning of Melodies**
  Partanen et al. (2013) had mothers listen repeatedly to a melody during their last trimester. After birth, EEG measurements showed their infants' brains responded more strongly to that specific melody than the brains of control infants, demonstrating musical learning in the womb.

- **2. Familiarity Drives Infant Preference**
  Plantinga and Trehub (2014) familiarized 6-month-olds with either consonant or dissonant music.
  Result: Infants familiarized with consonant music preferred consonant music.
  Crucially, infants familiarized with dissonant music preferred to listen to dissonant music.
  Conclusion: Preference was driven by familiarity, not an innate bias for consonance.





NotebookLM

Cross-Cultural Evidence & The Fluency Hypothesis.

If the preference for consonance is learned, it should not be universal.

McDermott et al. (2016) tested this by playing consonant and dissonant chords to people with varying exposure to Western music.

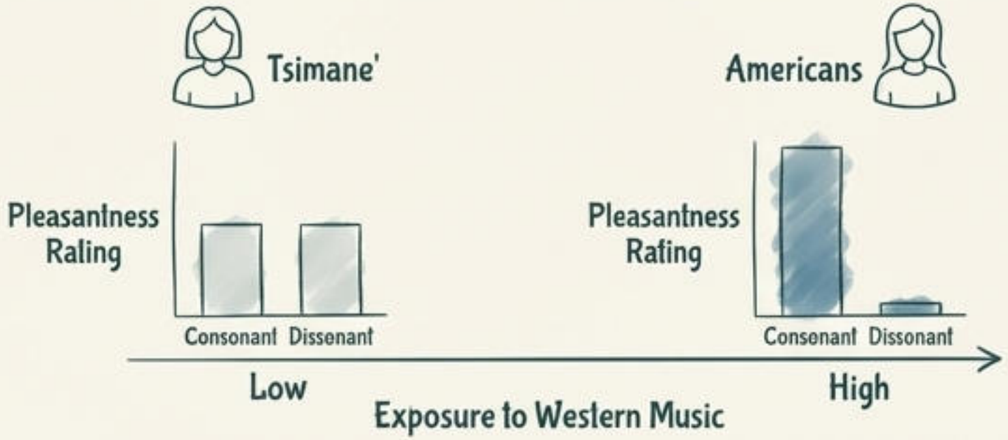
- **Participants**: Americans, Bolivian city dwellers, and the Tsimane', a remote Amazonian society with minimal exposure.
- **Results**: While Americans and Bolivians strongly preferred consonant chords, the Tsimane' showed no preference, rating both as equally pleasant.

The Processing Fluency Explanation

This leads to a unifying theory: **Processing Fluency**. We find stimuli that are easier for our brains to process more aesthetically pleasing.

- Through constant cultural exposure (lullabies, radio), our brains become highly efficient at processing consonant music. This fluency feels good.
- The Tsimane', lacking this lifelong exposure, process both types with similar effort and thus show no preference.

Our musical tastes may not be an innate echo of evolutionary harmony, but a beautiful reflection of the sounds our brains have learned to process with ease.



NotebookLM